Solar Fe II appendix

Sun · Fe II · downloadable appendix

Fe.json v1.390 · 30 Fe II products · 155 total

Solar Fe I anchor: 7.477 ± 0.052

HARPS · Reference Grade · Amarsi 3D-NLTE · n = 21. The selected Fe I product anchors this Solar appendix; products and engines are not averaged together.

Highlighted band products

near-UV

7.409 ± 0.081

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · n = 12

Uncertainty budget closed for six of eight rows (RYA-1226); two Fe I Kitt Peak/molecfit rows keep a legacy bar pending their microturbulence bracket. On the migrated rows most of the quoted systematic is a conservative profile-width term carried from a measurement in another band.

VIS

7.378 ± 0.118

Kitt Peak · Synth · 1D-LTE · Deep Grade · n = 23

Error-bar forest

How to read this forest. Each measured row is a separate abundance product for an instrument, holding, line pool and treatment; engines are never averaged together. Solid bars show statistical uncertainty from the line measurements; wireframes show systematic uncertainty from the product's adopted error budget, which does not average away with more lines. Green and gold regions mark the Asplund and Lodders literature values and uncertainties, not pass/fail gates. Grades describe the line pool's provenance and selection, not the quality of an engine. Reddish-orange mean-3D / Frankenstein rows are experimental and not adopted. Unmeasured model rows remain N/A.

Grades and model labels

Reference Grade. An externally published reference line pool, using that reference's line selection and oscillator strengths; the current Fe reference products use AGSS21. This is a provenance category, not a promise that every line has an individual primary-laboratory uncertainty.

Deep Grade. The corresponding laboratory-data pool above that depth gate. These stronger lines extend coverage but require particular care with saturation and blends.

1D-LTE uses a one-dimensional atmosphere with atomic populations in local thermodynamic equilibrium. **1D-NLTE** allows departures from that equilibrium; the named implementation supplies the correction or synthesis treatment.

Synthesis fits the spectral line profile. See Methodology for the full model, line-selection and uncertainty procedures.

Asplund, Amarsi & Grevesse 2021 7.46 ± 0.04 · Lodders+ 2025 7.49 ± 0.01

blue solid = statistical σ · wireframe = systematic σ · blue = regular model result · reddish orange = experimental Frankenstein / Gerber mean-3D; not adopted · green = Asplund; gold = Lodders · N/A = no product for this model

near-UV — Kitt Peak — Kurucz 2005 corrected Codex deep pool
solar_kpno_kurucz2005_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.14–7.79

EW · 1D-LTE no product		N/A

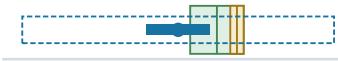
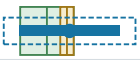
near-UV — Kitt Peak — Kurucz 2005 corrected Codex deep pool
solar_kpno_kurucz2005_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.14–7.79

Synth · 1D-LTE Deep Grade · n=31		7.535 ±0.044 stat ±0.087 syst
Synth · 1D-LTE · Gerber Deep Grade · n=31		7.535 ±0.044 stat ±0.088 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Deep Grade · n=16		7.656 ±0.061 stat ±0.100 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

near-UV — Kitt Peak — Kurucz 2005 corrected Reference pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.14–7.79

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=12		7.559 ±0.059 stat ±0.083 syst
Synth · 1D-LTE · Gerber Reference Grade · n=12		7.562 ±0.058 stat ±0.056 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=7		7.624 ±0.097 stat ±0.097 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

near-UV — Kitt Peak — 1984 Molecfit corrected Codex deep pool
solar_kpno_molecfit_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.14–7.79

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Deep Grade · n=12		7.402 ±0.047 stat ±0.233 syst
Synth · 1D-LTE · Gerber Deep Grade · n=29		7.435 ±0.051 stat ±0.063 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Deep Grade · n=16		7.494 ±0.075 stat ±0.099 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

near-UV — Kitt Peak — 1984 Molecfit corrected Reference pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.14–7.79

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=11		7.405 ±0.064 stat ±0.056 syst
Synth · 1D-LTE · Gerber Reference Grade · n=12		7.409 ±0.059 stat ±0.055 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=6		7.413 ±0.101 stat ±0.087 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

near-UV axis: 7.14 · 7.30 · 7.46 · 7.62 · 7.79 A(Fe) dex

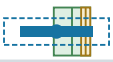
VIS — Kitt Peak — Kurucz 2005 corrected Codex deep pool
solar_kpno_kurucz2005_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.06–8.01

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Deep Grade · n=23		7.378 ±0.049 stat ±0.107 syst
Synth · 1D-LTE · Gerber Deep Grade · n=23		7.386 ±0.049 stat ±0.113 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Deep Grade · n=10		7.371 ±0.065 stat ±0.101 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

VIS — Kitt Peak — Kurucz 2005 corrected Reference pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.06–8.01

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=8		7.454 ±0.100 stat ±0.117 syst
Synth · 1D-LTE · Gerber Reference Grade · n=8		7.454 ±0.101 stat ±0.127 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=2		7.232 ±0.143 stat ±0.087 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

VIS — Kitt Peak — 1984 Molecfit corrected Codex deep pool solar_kpno_molecfit_corrected · DEEPGRADED (no graded product in this band) A(Fe) dex; axis 7.06–8.01

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Deep Grade · n=8		7.425 ±0.078 stat ±0.111 syst
Synth · 1D-LTE · Gerber Deep Grade · n=8		7.426 ±0.079 stat ±0.115 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Deep Grade · n=2		7.236 ±0.135 stat ±0.098 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

VIS — Kitt Peak — 1984 Molecfit corrected Reference pool solar_kpno_molecfit_corrected A(Fe) dex; axis 7.06–8.01

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=8		7.425 ±0.078 stat ±0.111 syst
Synth · 1D-LTE · Gerber Reference Grade · n=8		7.426 ±0.079 stat ±0.115 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=2		7.236 ±0.135 stat ±0.098 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

VIS — HARPS Codex deep pool solar_harps_molecfit_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.06–8.01

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Deep Grade · n=23		7.456 ±0.072 stat ±0.117 syst
Synth · 1D-LTE · Gerber Deep Grade · n=23		7.457 ±0.073 stat ±0.102 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Deep Grade · n=10		7.444 ±0.075 stat ±0.126 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

VIS — HARPS Reference pool solar_harps_molecfit_corrected
A(Fe) dex; axis 7.06–8.01

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Reference Grade · n=8		7.848 ±0.127 stat ±0.113 syst
Synth · 1D-LTE · Gerber Reference Grade · n=8		7.850 ±0.128 stat ±0.105 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Reference Grade · n=2		7.446 ±0.049 stat ±0.110 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

VIS axis: 7.06 · 7.30 · 7.53 · 7.77 · 8.01 A(Fe) dex

Read all product values, including alternate line sets and experiments

HARPS · Synth · 1D-LTE · Deep Grade · solar_harps_molecfit_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.456 ± 0.138 reported total; n=23; ξ MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_harps_molecfit_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.444 ± 0.146 reported total; n=10; ξ MEASURED

HARPS · Synth · 1D-LTE · Gerber · Deep Grade · solar_harps_molecfit_corrected · DEEPGRADED · SYNTH · synth-1D-LTE-gerber: 7.457 ± 0.125 reported total; n=23; ξ MEASURED

HARPS · Synth · 1D-LTE · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.848 ± 0.170 reported total; n=8; ξ MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.446 ± 0.121 reported total; n=2; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

HARPS · Synth · 1D-LTE · Gerber · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.850 ± 0.165 reported total; n=8; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.378 ± 0.118 reported total; n=23; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.371 ± 0.120 reported total; n=10; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · synth-1D-LTE-gerber: 7.386 ± 0.123 reported total; n=23; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · 1D-LTE: 7.454 ± 0.153 reported total; n=8; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A: 7.232 ± 0.168 reported total; n=2; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.454 ± 0.162 reported total; n=8; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_molecfit_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.425 ± 0.136 reported total; n=8; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_molecfit_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.236 ± 0.167 reported total; n=2; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Deep Grade · solar_kpno_molecfit_corrected · DEEPGRADED · SYNTH · synth-1D-LTE-gerber: 7.426 ± 0.140 reported total; n=8; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.425 ± 0.136 reported total; n=8; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.236 ± 0.167 reported total; n=2; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.426 ± 0.140 reported total; n=8; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.535 ± 0.098 reported total; n=31; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.656 ± 0.117 reported total; n=16; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · synth-1D-LTE-gerber: 7.535 ± 0.099 reported total; n=31; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · 1D-LTE: 7.559 ± 0.102 reported total; n=12; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A: 7.624 ± 0.138 reported total; n=7; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.562 ± 0.081 reported total; n=12; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_molecfit_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.402 ± 0.238 reported total; n=12; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_molecfit_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.494 ± 0.124 reported total; n=16; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Deep Grade · solar_kpno_molecfit_corrected · DEEPGRADED · SYNTH · synth-1D-LTE-gerber: 7.435 ± 0.081 reported total; n=29; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.405 ± 0.085 reported total; n=11; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.413 ± 0.133 reported total; n=6; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.409 ± 0.081 reported total; n=12; ξ MEASURED

Solar iron context

Fe II traces singly ionized iron and is retained as an ionization and consistency diagnostic alongside Fe I, not an independent headline to average with it. The current resolved pool contains strong, saturation-sensitive lines and documented blend exclusions; the diagnostics below distinguish those limitations from model and holding differences.

Near-UV opacity — report and noted

Molecular opacity INCLUDED — OH A-X, NH, CH and CN violet, 6,223 VALD lines as Turbospectrum lists with the dominant isotopologue (RYA-1207). The band previously carried none: every molecular list iSpec ships begins at 400 nm. Measured PAIRED against a molecules-off run on the same code and pool, the lever is -0.047..-0.055 dex on Fe I and -0.276..-0.332 on Fe II, with chi2r median 81.8 -> 67.4 and control windows containing no molecular line bit-identical. RYA-1209 additionally curated out Fe I 3721.272 (systematic:K07 — gf attributed to a VALD3 reference that publishes no per-line uncertainty, no NIST accuracy class, so it never met the DEEPGRADED tier's lab-gf definition); the Fe I pool is 57 candidates and the budget clears at gf rung 3 (cited lab), 55/55 GF-LAB, RMS 0.0509 dex. RESIDUAL DEFICIT: the correction still UNDER-corrects — RYA-1204 measured synth/obs above 1 on 24 of 40 windows after it — so the band remains short of opacity by an amount not yet measured on this pool, and these values are a floor rather than a converged answer. sigma_syst is unchanged at 0.1122 because it is dominated by the 0.10 dex pseudo-continuum term, not by gf: the true continuum is never observed here. Not adopted for the headline. The headline is the per-band best-available product, selected from this feed by the publication generator rather than fixed as a constant: RYA-850 promoted the graded laboratory-gf pool to the PRIMARY reported value, and RYA-819 retired the earlier constructed anchor after showing its Magic-2013 correction wrong in both magnitude and shape. RYA-161: nothing calibrated to a target.

Frankenstein — experimental, pending the Bride

2 experimental products remain available in the Fe I product values and downloads.

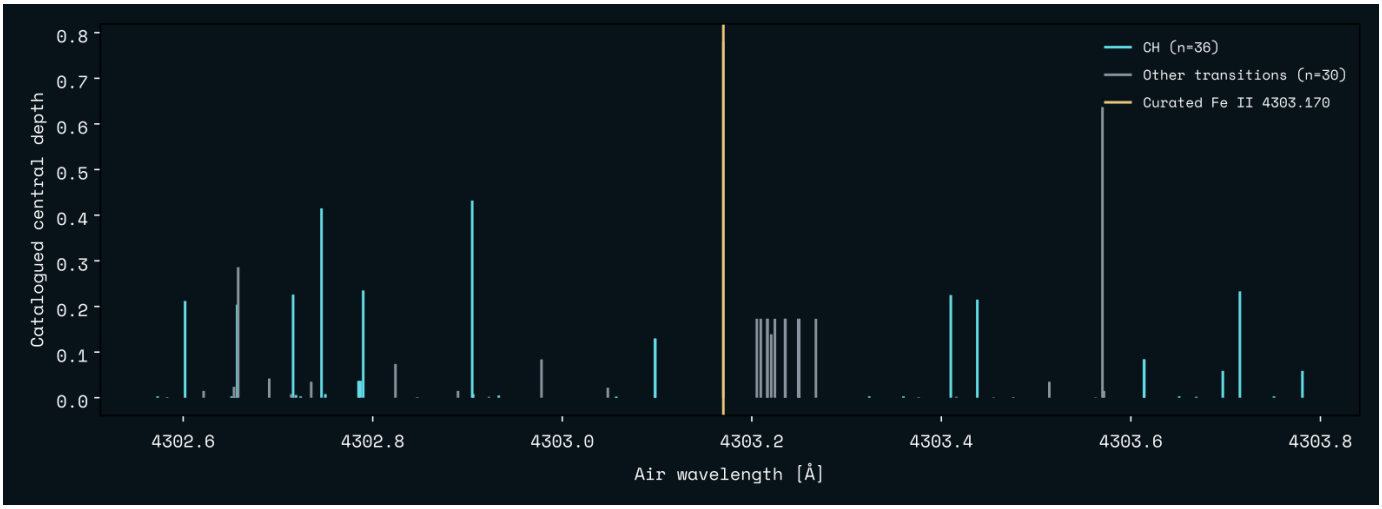
RYA-1197 Part A: the Frankenstein mean-<3D> route is HELD FROM HEADLINE publication until the Bride validates it. It inflates A(Fe) by roughly +0.10 dex against the 1D siblings and exists on VIS/KP only, so it is not a comparable measurement of the same quantity. It stays in the feed deliberately: the LTE/NLTE mean-3D PAIR is the NLTE differential on ONE atmosphere (RYA-542), which is the diagnostic it is for. Not a headline value.

Saturation characterization

Committed proof census: 120 line/holding records; 100% have observed depth > 0.70. Median observed log10(EW[mÅ]/λ[Å]) = -1.252. In consistent Å units this is -4.252. These diagnostics characterize the audited saturated pool, not every newly added product.

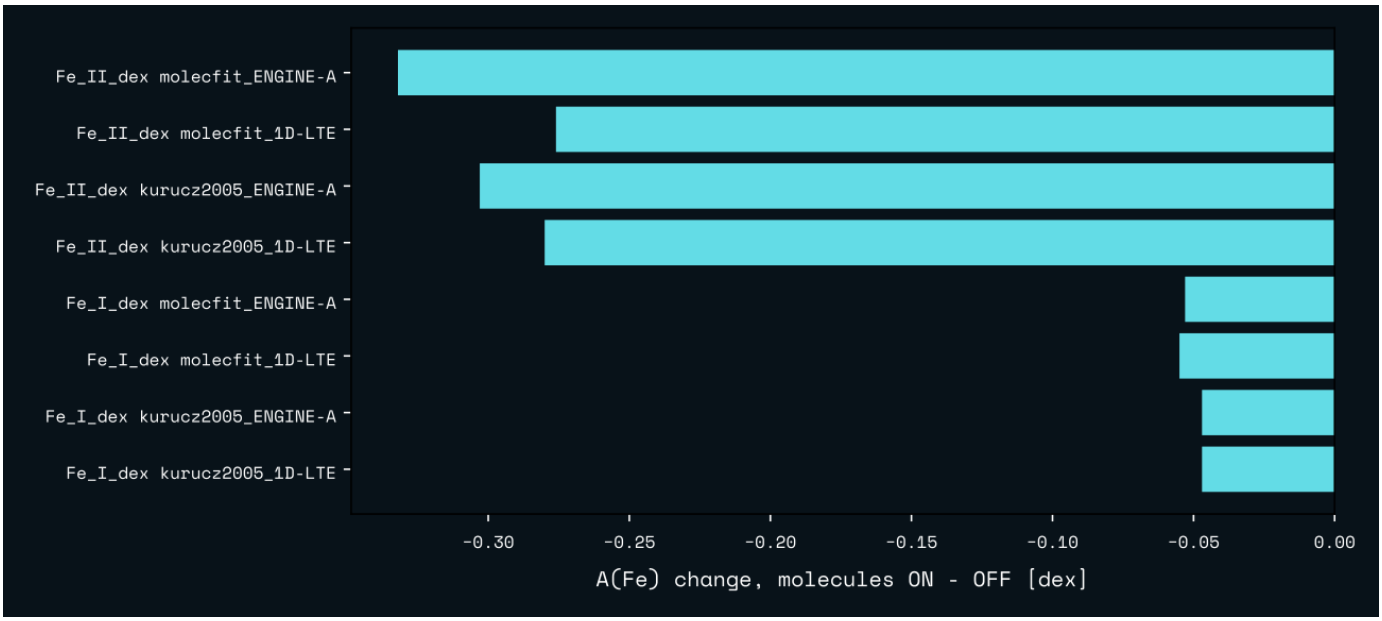
Problem-line and model diagnostics

CH G-band artifact



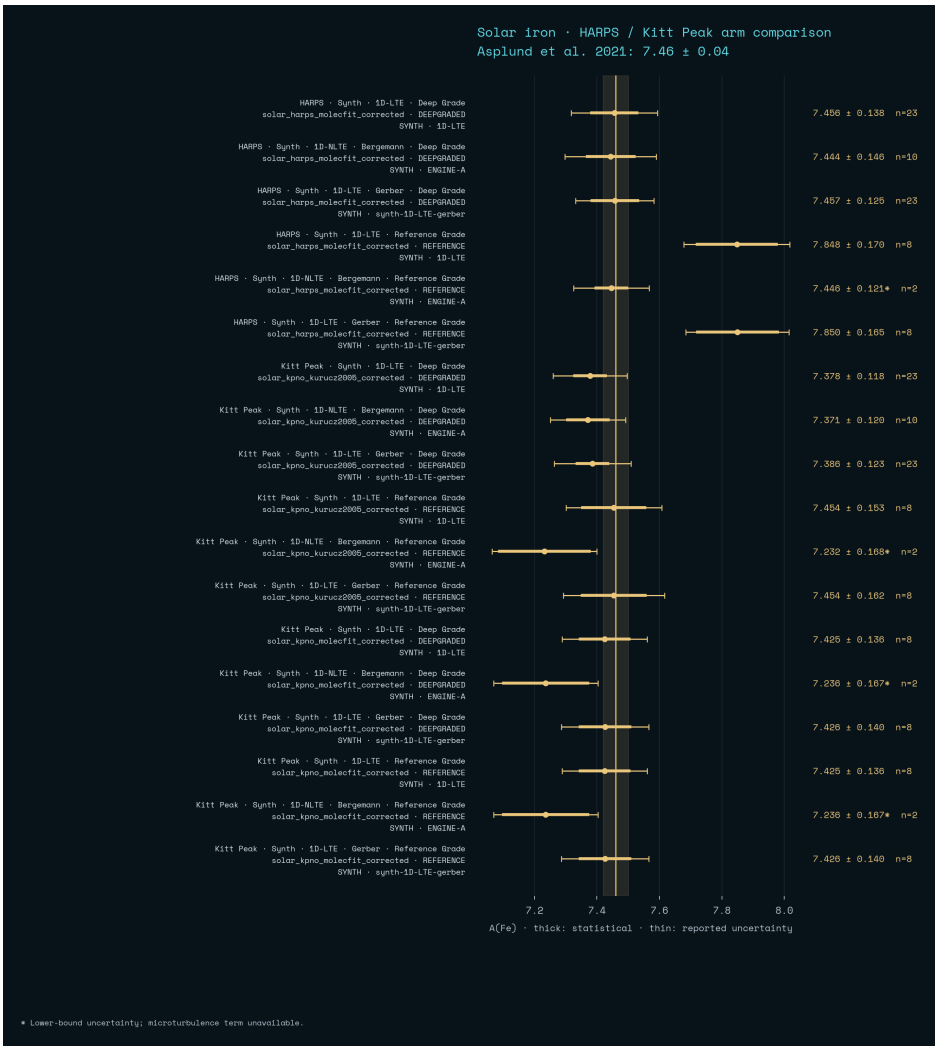
CH G-band blend artifact — the synthesis list cannot represent the dominant absorber · A = 10.0 (+2.5 dex). RYA-1191 blend audit: 66 catalogued transitions in the 0.62 A fit window, top contaminant CH I at 43.8% of the summed depth against Fe II's 11.1%. The VIS synthesis list GESv6_atom_hfs_iso carries 206 species and NOT ONE molecular, so the fit had nowhere to put the CH absorption except Fe II. RYA-515 independently counted 36 of 65 transitions as CH. The plot shows the committed transition census in the proof generator's fit window; it is not a synthetic spectrum. Source evidence

Near-UV molecular opacity



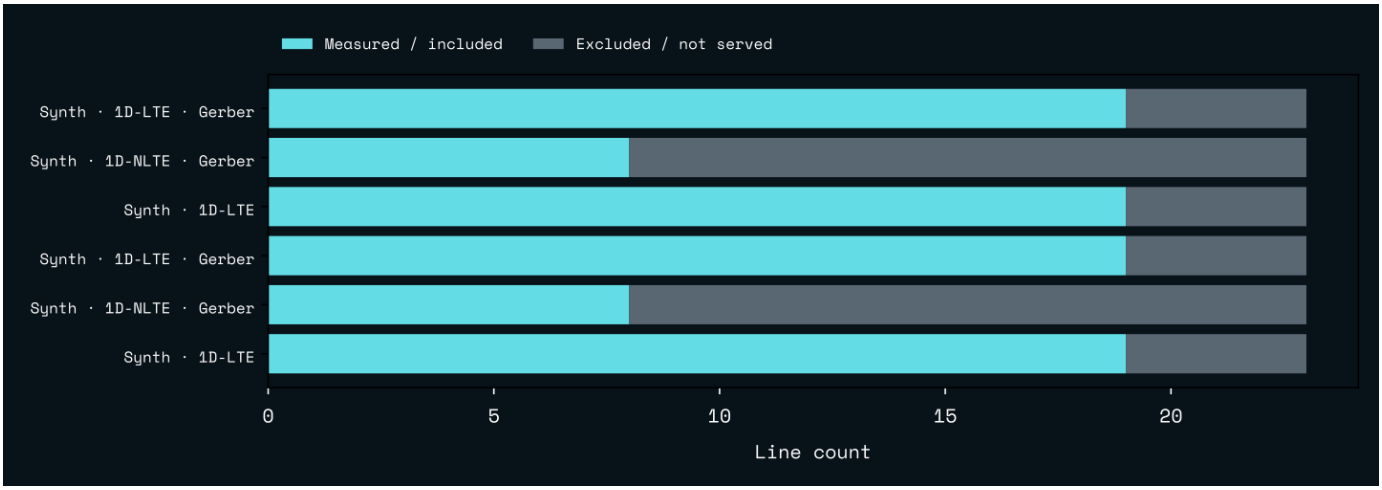
molecules OFF vs ON on THE SAME CODE and the same pool. Differencing against the published values instead would have measured the molecular lever PLUS every code change since the feed was written on Sept 3 -- which is about -0.035 dex of the total. Source evidence

HARPS arm offset



Current Fe II VIS products, separated by holding and engine. The arm offset persists after the curated CH artifact was removed. Source evidence

CRIRES+ H model coverage



Included and excluded line counts from each current product. Excluded rows are not automatically failed fits; the per-line reason distinguishes unavailable NLTE coverage. Source evidence

Download the evidence

809 graded per-line measurements exported. Evidence for 137 graded products is missing or does not reproduce the feed; 0 exported lines lack an unambiguous gf match. The download is explicitly incomplete until those upstream records are supplied.

Graded per-line abundances (CSV)

All product fields (CSV; nested fields preserved as JSON)

Per-product line-evidence coverage (CSV)

Source Fe.json

Reproducibility

Generator scripts/generate_fe_publication.py v1.0.0
Science source commit c93d65e62c04a990f59dfb57bdb0b41f87966c58
Fe.json v1.390 · feed timestamp 2026-10-19T05:21:12Z
Generated 2026-10-19T05:21:38+00:00
Feed SHA-256 1308bb0ac396bbd99db4a12ee2572193bcf54318bda29aa291ea0f8303819591

Refreshed product tracker · Build manifest · Fe I appendix

References

Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision (doi: 10.1051/0004-6361/202140445). AGSS21 Solar reference and the green literature comparison; Table A.2 defines the Reference Grade Fe I line set.

Lodders, K.; Bergemann, M.; Palme, H. (2025). Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites (doi: 10.1007/s11214-025-01146-w). Table 6 present-day Solar composition, shown as the gold literature comparison.

Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum (doi: 10.1051/0004-6361/202243673). The Gerber Turbospectrum non-LTE synthesis implementation named in the model rows.

Den Hartog, E. A.; Lawler, J. E.; Sneden, C.; Cowan, J. J.; Brukhovesky, A. (2019). Atomic Transition Probabilities for UV and Blue Lines of Fe II and Abundance Determinations in the Photospheres of the Sun and Metal-poor Star HD 84937 (doi: 10.3847/1538-4365/ab322e). Laboratory transition probabilities used in the resolved Fe II per-line evidence. Coverage and unresolved products are listed in the downloads.